

A. J. A. JUBAYER TALUKDER

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 github.com/JubayerONROB

EDUCATION

Bangladesh University of Engineering and Technology (BUET)

Bachelor of Science in Electrical and Electronics Engineering (EEE)

Major in Communication and Signal Processing (CSP)

Dhaka, Bangladesh

Jan. 2022 – June 2026

CGPA: 3.83 (Out of 4.00)

THESIS AND ONGOING RESEARCH

Proactive Conversation Assistant: LLM Pipeline  | *NLP, Edge AI*

Feb. 2025 – Present

- Developed a real-time wearable assistant that decides *when to interrupt* a conversation and *what to say*, using a two-stage LLM pipeline evaluated on 28,058 decision points; reduced unnecessary generator calls by $7.62\times$ and achieved up to $4.39\times$ lower wall-clock time with an 8B generator.
- Analyzed factors affecting hint quality and found persistent user-memory context to be the strongest contributor, with a 24.7-point drop in exact match when removed; designed a compact 1B-parameter decoder that shares one adapter between interruption detection and response generation, reducing resident weights to 1.05 GB and showing $1.3\text{--}1.8\times$ lower compute than the two-model pipeline.

PROJECTS

Autonomous Rescue Drone for Locating Survivors  | *Computer Vision, AI, UAV*

Sep. – Dec. 2024

- Developed an AI-powered drone for search and rescue; implemented real-time human detection using RGB cameras and OpenCV.
- Integrated GPS-based autonomous flight control for structured area coverage.

Ultrasonic Obstacle Mapping Bot  | *Embedded Systems, Robotics, IoT*

August 2025

- Built a low-cost (under \$50) autonomous mapping robot: an HC-SR04 ultrasonic sensor on a servo performs 180° scans while a NodeMCU ESP8266 streams the resulting indoor obstacle map to a live web interface.
- Implemented polar-to-Cartesian coordinate conversion for real-time map plotting and closed-loop obstacle avoidance driven by an L298N motor driver, dual DC gear motors, and wheel encoders.

Hybrid Router: Two-Stage Local-First LLM Agent  | *LLM Routing, Docker, Python*

July 2026

- Built a token-efficient LLM routing agent for AMD Hackathon ACT II (Track 1); a grammar-constrained local GGUF classifier (Qwen2.5-3B, Q4_K_M) resolves shallow tasks at zero API cost, escalating only high-difficulty queries across 8 intent categories to a Fireworks remote backend.
- Dockerized the full inference pipeline for a CPU-only grading VM (4 GB RAM, no GPU, linux/amd64) with env-driven model selection, 25s per-request timeout, graceful local fallback, and an offline pytest evaluation harness ensuring zero hardcoded values.

16-bit Binary Square Rooter  | *VLSI, Verilog, SystemVerilog, Cadence EDA*

Jan. – Apr. 2026

- Designed a multiplier-free 16-bit integer square-root module in Verilog (shift-and-subtract, 8-iteration unrolled); achieved single-cycle throughput at 100 MHz and 100% functional coverage across directed, layered-SV, and covergroup testbenches.
- Executed full RTL-to-GDS flow (Cadence Genus and Innovus, 45 nm GPDK045): 197 cells, 347 sq. um area, DRC-clean layout with zero timing violations – setup WNS $+0.264$ ns, hold WNS $+0.014$ ns.

Deepfake Detection using CLIP-ViT  | *OpenCLIP, ViT, PyTorch*

Oct. 2024 – Jan. 2025

- Developed a deepfake detection model using OpenCLIP-ViT; achieved 98% training and 97% validation accuracy.
- Tested on IEEE SP Cup 2025 dataset (43k real, 219k fake images) with strict mini-batch training.

Brain Tumor Detection on MRI Images  | *Digital Image Processing, MATLAB*

Mar. 2024

- Developed a CAD system for automated brain tumor detection; improved MRI image clarity via advanced processing techniques.
- Achieved high sensitivity and specificity in tumor detection on clinical MRI data.

RESEARCH INTERESTS

LLMs for Automated Hardware Design & EDA | AI Hardware and Edge Computing | Model Optimization and Efficient Inference | Embedded and Wearable AI | Robotics and Autonomous Systems | Computer Vision | Natural Language Processing

RELEVANT COURSEWORK

Core Foundations: Linear Algebra | Probability and Statistics | Electrical Circuits | Energy Conversion | Electrical Service Design | Computer Programming

Signals and Communications: Continuous Signals and Linear Systems | Digital Signal Processing | Random Signals and Processes | Communication Systems | Wireless Communication | Radar and Satellite Communication | Optical Communications

Intelligent Systems and AI: AI and ML | Medical Image Processing | Microprocessors and Embedded Systems

Electronics and Hardware: Solid State Devices | Electronic Circuits | Digital Electronics | VLSI Circuits and Design | Power Electronics | Power System | Control System

TECHNICAL SKILLS

Languages: Python, SQL, C/C++, Verilog, SystemVerilog, HTML/CSS, Matlab, Bash, LaTeX

Frameworks and Libraries: PyTorch, TensorFlow, Keras, OpenCV, Hugging Face, Unsloth, llama.cpp, scikit-learn, Pandas, NumPy

Specializations: Large Language Models (LLMs), NLP, LLM Routing, Computer Vision, UAV, VLSI Design, RTL-to-GDS, PCB Design

Developer Tools: Docker, Cadence Genus, Cadence Innovus, Cadence Xcelium, Visual Studio, PyCharm, Kaggle, Colab, SolidWorks, LangChain, PSS/E, PSAF, Intel Quartus, Raspberry Pi, Arduino, Git, EasyEDA Pro

VOLUNTEER EXPERIENCE

BUET Robotics Society	March 2023 – June 2026
<i>Head of Design Team, Deputy Head of Event Management</i>	<i>Dhaka, Bangladesh</i>
• Led design and event teams, coordinating robotics projects and managing key events.	
IEEE BUET Student Branch	January 2024 – June 2026
<i>Chairperson (SPS), Vice Chairperson (IAS), Executive Member</i>	<i>Dhaka, Bangladesh</i>
• Coordinated industry-academia collaborations and organized technical workshops and events.	

ACHIEVEMENTS

1st Runner Up (Int'l), IEEE-IAS Humanitarian Contest 	IEEE IAS Annual Meeting 2025
1st Runner Up, IEEE-WIE Robotics for Climate Change	WIE BD Summit 2024
2nd Runner Up, Project Showcasing	Intra BUET Robo Challenge